

Apple Peel Unfolding of Archimedean and Catalan Solids

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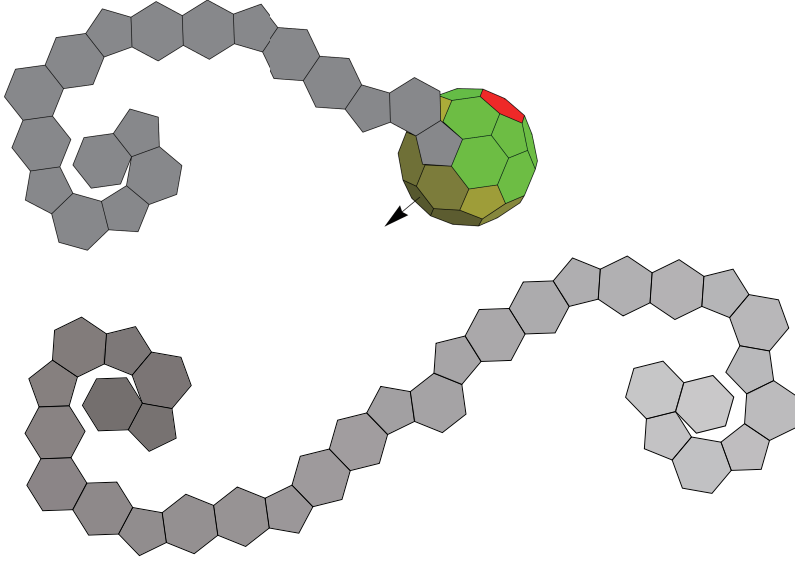
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ABSTRACT

We considered a new kind of the methods for making polyhedron nets referred to as “apple peeling unfolding”: drawing the nets as if we peel apple skins. We define the apple peel unfolding strictly and code a program that calculate the order of the selection of the polyhedral faces in accordance with the definition. The program determines given polyhedron as peelable (possible to be peeled) or not. We classify Archimedean solids and their dual polyhedrons (Catalan solids) into perfect (always peelable), possible (peelable for restricted cases), or impossible. The results show that four of Archimedean solids and five of Catalan solids are perfect and three of Archimedean ones and three of Catalan ones are possible.



KEYWORDS

Polyhedron net, apple peeling; Archimedean solids; Catalan solids

1. Introduction

Unfolding of polyhedral surfaces is a well-known subject as a method to obtain polyhedron nets. Peeling of apple, on the other hand, is a quite common procedure in our daily lives. Therefore, considering the combination of these two subjects is expected to have fruitful results. However, there are no systematic treatment of the unfolding yet. For this reason, we discuss the unfolding referred to as “apple-peel unfoldings” in this article.

There are some considerations of polyhedron similar to peeling apples. Zipper Unfolding (E. D. Demarie et al. , 2010) a polyhedron net based on the Hamiltonian path of a polyhedron. (Demaineは何をやった?) As described in Section 4, apple peel unfolding corresponds to a special case of it. Spiral Unfolding(O’Rourke, J. , 2015) expands surfaces of polyhedrons by cutting its faces spirally. The unfoldings considered in this study can be regarded as the special cases of the method: the cuttings are restricted with only the edges.

Apple peeling of regular polyhedron has already examined by some researchers[ref, ref]. This procedure can be regarded as sequentially and spirally unfolding of outermost layer of three dimensional objects with respect to an axis. The results of apple peel

unfolding for Platonic solid are reported on and produced as commercial materials (Kaino, K. , 2021).

Our objectives of this research are to define the "apple-peeling of a polyhedron" strictly and to obtain the results for semi-regular polyhedrons automatically. The former objective is necessary to extend the topic to higher-dimensional polytopes (e.g., six regular polytopes in the four-dimensional space) although we focus on only three-dimensional polyhedrons in this study. It should be noted that the apple-peel unfolding is related with graph theory because peelings of polyhedrons belong to a problem of the Hamiltonian path of the dual polyhedrons. However, the solution cannot be obtained from their planar graphs directly due to the existence of their axis.

The remaining part of this article consists of three parts: the definition and the algorithm of apple peel unfolding are described in section 2. The results of numerical simulations of the unfoldings of semi-regular polyhedrons, Archimedean and Catalan solids, are explained in section 3. Finally, we discuss the results in section 4.

2. Methods

2.1. Definition of apple peeling unfolding and algorithm for numerical simulation

We define apple peel unfolding of a three-dimensional polyhedron in the three-dimensional space R^3 in the followings. It is an unfolding of polyhedral faces according to a fixed axis that contains the centroid of the polyhedron. And, it is a sequential selection of all the faces clock-wisely (or anti-clock-wisely) around the axis from top to bottom. The direction of the selection corresponds to the dominant hand of a handler hereafter. We assume that the handler is a right-handed person.

Assume that a three-dimensional polytope P is located on three-dimensional space R^3 . A couple of neighboring faces F_1 and F_2 is chosen arbitrary and the apple-peeling of the facets starts from F_1 , and choose F_2 as the next to be peeled. The Cartesian coordinates $O-x_1x_2x_3$ are set in order that the origin is located on the centroid of P and the x_3 -axis contains the centroid of F_1 . Then, the axis of peeling of the polytope is defined as x_3 -axis that contains both the centroids of P and F_1 . The coordinates of the

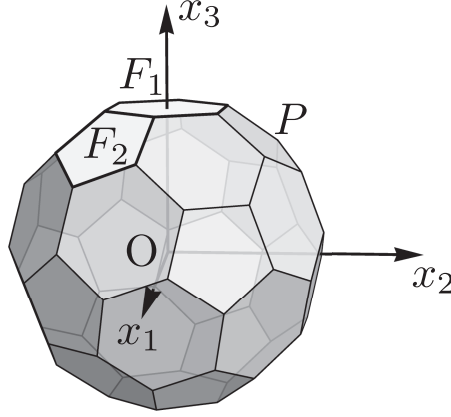


Figure 1. Definition of coordinates in three-dimensional space.

centroid of F_1 is set as $(0, 0, x_{3,1})$ automatically and we set the direction of the x_3 -axis according to $x_{3,1} > 0$. Therefore, we regard that F_1 is set on the top. As an example of above definition, we consider the case of the three-dimensional space. Figure 1 shows the schematic illustration of the case of a truncated icosahedron. We select a couple of neighboring faces F_1 and F_2 . The coordinates is $O-x_1x_2x_3$ and the centroid of P is located at the origin. Then, the axis of peeling is x_3 . We set the coordinates as described above. Second, we select one of the neighboring faces, candidates, according the following priority. If the candidate is only one, we use it automatically. If there are multiple candidates, we select the face of which the centroid is at the highest position among the faces on left side. If there is no candidates on the left side, we select the polygon of which centroid is located on the lowest among the polygons on right side. Finally, we continue the selection until all the faces are selected or there is no unselected neighbor.

Figure 2 shows an example of the intermediate state of the unfolding procedure of a truncated dodecahedron. Gray and green polygons show the unfolded and not unfolded faces, respectively. We add the original polyhedron with yellow and represent the finally selected face with red. The image illustrates that the peeling was started from a hexagon, a handler chose a pentagon as the second polygon, and the peeling continued till the selection of the eighteenth face.

The term *peelable* is defined as a situation all faces are selected when we apply the algorithm below to a polyhedron. It is determined by the sequential selection of two

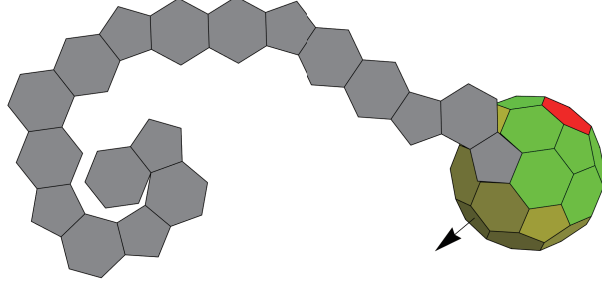


Figure 2. An example of intermediate state of apple peel unfolding of truncated icosahedron.

neighboring faces. By examining all two sequential selection of the neighboring faces, the nature of a polyhedron is determined either peelable or not.

We refer to a possibility of apple peeling unfolding of a polyhedron as *peelability* in the followings. The peelability can be classified into three types: *perfect*, *impossible*, and *possible* polyhedrons. Perfect polyhedrons can be peeled from any neighboring two faces. Impossible polyhedrons are polyhedrons which cannot be peeled from any pairs of neighboring two faces.

The algorithm for numerical simulation is described as follows: input data consist of three sets: (1) vertex coordinates $V = \{v_1, v_2, \dots, v_m\}$, where m denotes the number of vertices of the polyhedron and $v_i = \{x_{1,i}, x_{2,i}, x_{3,i}\}$, (2) vertex indices of all faces $\{f_1, f_2, \dots, f_n\}$, where n denotes the number of faces and f_i consists of k integers if the i -th polygon is a k -gon, (3) two neighboring face Indexes $\{F_1, F_2\}$ to be started. The face F_2 must be the neighbor of the first face F_1 . Its output is an order of face index, $\{F_1, F_2, \dots, F_j\}$, and the unfolding is regarded to be finished successfully if $j = n$. We refer to the result in the following as the selection order.

1. Allocate a target polyhedron in the 3D Cartesian coordinate space in order for that the centroid of the polyhedron is set at the origin of the coordinates.
2. Set the counting variable $k = 2$, and rotate the polyhedron to be set the centroid of the first polygon F_1 on the top; this means the rotation of the polyhedron to be set the position vector of the centroid of F_1 on the z -axis.
3. Select a face F_{k+1} among the adjacent faces of F_k based on the following priority:
 - (a) a. Select the remained face automatically as F_{i+1} if it is the only unselected neighbor.

Algorithm 1 peeling

Require: F_2 must be one of the neighbors of F_1 .

```
1: function APPLE_PEELING( $V = \{v_1, v_2, \dots, v_m\}$ ,  $F = \{f_1, f_2, \dots, f_n\}$ ,  $I = \{F_1, F_2\}$ )
2:    $k \leftarrow 2$ 
3:    $l \leftarrow$  a list of unselected neighbors of  $F_k$ 
4:    $i \leftarrow$  number of the elements of  $l$ 
5:   repeat
6:     if  $i = 1$  then
7:        $F_{k+1} \leftarrow$  the face number representing the unselected neighbor face
8:     else if  $i > 1$  then
9:        $q \leftarrow$  number of centroids on the left half
10:      if  $q > 0$  there are somethings then
11:         $F_{k+1} \leftarrow$  the face number of the face of which  $z$ -coordinate of the
        centroid is the largest
12:      else
13:         $F_{k+1} \leftarrow$  the face number of the face of which  $z$ -coordinate of the
        centroid is the smallest
14:      end if
15:    end if
16:     $k \leftarrow k + 1$ 
17:     $l \leftarrow$  a list of unselected neighbors of  $F_k$ 
18:     $i \leftarrow$  number of the elements of  $l$ 
19:  until  $i > 0$ 
20:  return  $\{F_1, F_2, \dots, F_k\}$ 
21: end function
```

(b) b. First, divide the Cartesian space into two part by the plane composed by the z -axis and the centroid of F_k . Second, list the adjacent polygons of which centroids belong to the left-half space. Finally, select the face which has the lartgest value of the z -coordinate as F_{k+1} ,

(c) c. To select the face as F_{n+1} of which z -coordinate is the highest among the neighboring faces on the left half-space. If there is no neighboring face on the left half-space, select the neighbor of the lowest z -coordinate.

4. Go back to #3 unless all the faces are selected or next face has no neighboring face.

2.2. Visualization

We used three types of visualizations in the followings: a) three dimensional representation to display the sequential selection, b) polyhedron net, and c) planar graph. We have already shown the example of the three-dimensional representation in Fig. 2 with the description. In addition to the description, we used red polygons to represent the remained ones during the peeling. This means that existence of red polygons indicate that the peeling is failed. Polyhedron net is used to display the results of the unfolding when it finishes successfully. Planar graph is used to illustrate whole the trace of the selection.

The three-dimensional representation and the polyhedron net use the same procedure: the reptation of rotations of polygons. In the case representing the intermediate state of peeling in the three dimensional space, we rotate the polygons F_k ($k = 1, 2, 3, \dots, i$) with respect to the edge shared by F_i and F_{i+1} simultaneously in order that the angle between the two faces being flat. We obtain the polyhedron nets by repeating the rotation from $i = 2$ to $i = n - 1$ continuously where n denotes the number of elements of the order. It should be noted that some disconnections were observed in the cases of some polyhedron nets for Catalan solid due to numerical errors. We corrected such disconnections manually if necessary.

Insides of polyhedron are faced to us.

One is to draw the polyhedron net based on the result. A Gray scale represents the order; the color becomes lighter from start to finish.

Planar graph representation is useful especially in the cases of uniform polyhedrons because the symmetry properties of the polyhedrons almost conserved.

The Tutte method is used for graph drawing.

3. Results

3.1. Numerical Simulations

We carried out numerical simulations by using Mathematica®. The basic data of polyhedrons such as vertex coordinates, vertex numbers of each face, are obtained

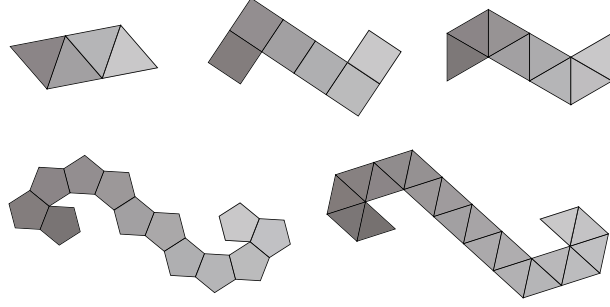


Figure 3. Polyhedron nets of Platonic solids obtained from apple peel unfolding.

from the library of Mathematica. Although MathematicaTM enables us to manipulate exact formula (e.g., π and $\sqrt{3}$), we used approximated values to minimize the execution time because the simulation using exact formula took unrealistic time. Consequently, the results were sometimes affected by the numerical errors occurred by the difference between exact and approximated values. We tried all the possible combinations of $\{F_1, F_2\}$ as the initial values of the simulation and summarized the results by the comparisons of their polyhedron nets.

3.2. Platonic Solids

All the Platonic solids were classified to perfect as shown in Fig. 3. The results has already been known well(Kaino, K. , 2021). For dodecahedron, we obtained only one type of the polyhedron nets although Kaino listed two types of polyhedron nets. This means that only the polyhedron net in Fig. 3 matches our definition of apple peel unfolding.

3.3. Archimedean Solids

The results for Archimedean and Catalan solids are summarized in Fig. 4 and Table 1. We represent the types of the Archimedean and Catalan solids by using the curly brackets and square brackets, respectively. We used the notations as $\{n_1, n_2, \dots\}$ and $[n_1, n_2, \dots]$ to refer Archimedean and Catalan solids. The former notation indicates each vertex of the polyhedron consists of n_1 -gon, n_2 -gon, $\dots$, and the latter one does that each face of the polyhedron consists of vertices of degrees $n_1, n_2, \dots$. Therefore, a couple of polyhedrons having a dual relation is represented with the same num-

Archimedean Solids

Perfect

$\{4,6,10\}$



$\{5,6,6\}$



$\{4,6,6\}$

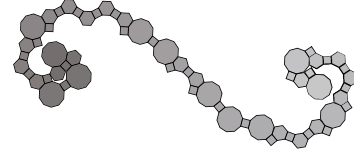


Possible

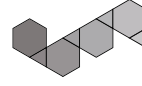
$\{3,3,3,3,4\}$



$\{4,6,8\}$



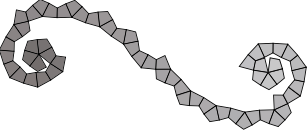
$\{3,6,6\}$



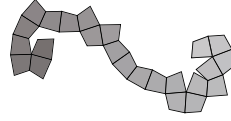
Catalan Solids

Perfect

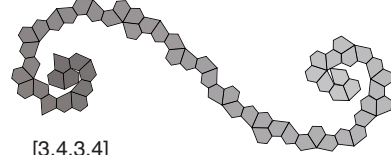
$[3,4,5,4]$



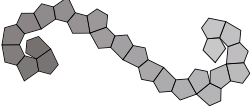
$[3,4,4,4]$



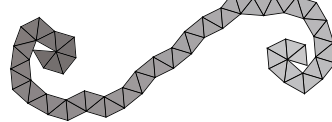
$[3,3,3,3,5]$



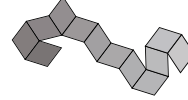
$[3,3,3,3,4]$



$[5,6,6]$

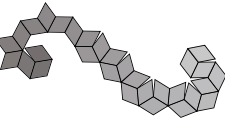


$[3,4,3,4]$

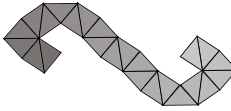


Possible

$[3,5,3,5]$



$[4,6,6]$



$[3,6,6]$

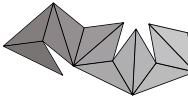


Figure 4. Examples of polyhedron nets of all succeeded polyhedrons.

ber combinations with different type of brackets. Figure 4 illustrates the examples of the polyhedron nets that the unfolding has finished successfully for all peelable polyhedrons. For Archimedean solids, the numbers of the perfect, possible, and impossible polyhedrons are four, three, and six, respectively. On the other hands, those for Catalan solids are five, three, and five, respectively. There is no apparent relation of the results of Archimedean solids and their duals. Therefore, we conclude that the symmetry property of the faces or vertices does not the critical role for peelability.

The failure of the peelings can be classified to two types. One is the cases that all neighbors have already selected (termination) and the other is the unchosen face(s) have found when the peeling process almost ends (isolation). Although most of the termination cases were found at a triangle,

Table 1. Summary of numerical simulations. Bold letters represent that their whole results are described in the paper.

Name	Index	Peelability		Hamiltonian path of Catalan solids
		Archimedean	Dual(Catalan)	
Cuboctahedron	$\{3,4,3,4\}$	Impossible	Perfect	n.a.
Great Rhombicosidodecahedron	$\{4,6,10\}$	Possible	Impossible	available
Great Rhombicuboctahedron	$\{4,6,8\}$	Perfect	Impossible	available
Icosidodecahedron	$\{3,5,3,5\}$	Impossible	Possible	n.a.
Small Rhombicosidodecahedron	$\{3,4,5,4\}$	Impossible	Perfect	n.a.
Deltoidal Icositetrahedron (Rombicuboctahedron)	$\{3,4,4,4\}$	Impossible	Perfect	n.a.
Snub Cube	$\{3,3,3,3,4\}$	Possible	Perfect	available
Snub Dodecahedron	$\{3,3,3,3,5\}$	Impossible	Perfect	available
Truncated Cube	$\{3,8,8\}$	Impossible	Impossible	n.a.
Truncated Dodecahedron	$\{3,10,10\}$	Impossible	Impossible	n.a.
Truncated Icosahedron	$\{5,6,6\}$	Perfect	Perfect	available
Truncated Octahedron	$\{4,6,6\}$	Perfect	Possible	available
Truncated Tetrahedron	$\{3,6,6\}$	Possible	Possible	available

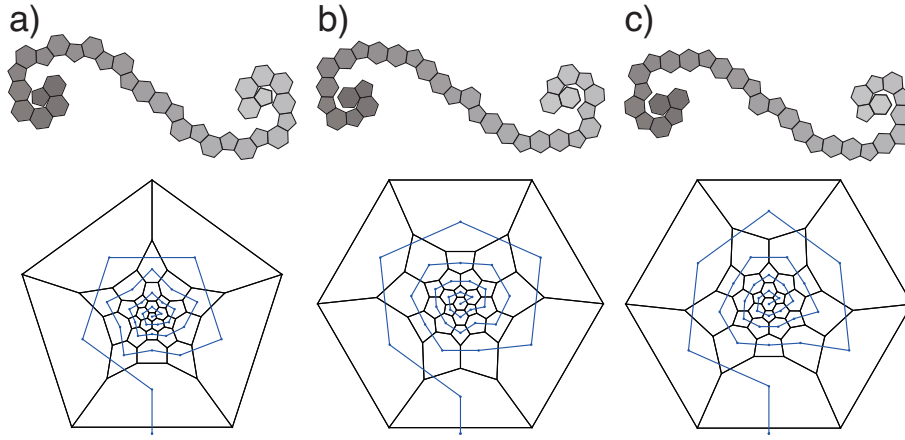


Figure 5. Polyhedron nets of a truncated icosahedron $\{5, 6, 6\}$.

the cases were occurred at other types of n -gons. Examples of “termination” and “isolation”.

The polyhedron nets of a truncated icosahedron $\{5, 6, 6\}$ were summarized as three patterns shown in Fig. 5. The polyhedron was perfect: it can be peeled starting from arbitrary two neighboring faces. The results of the polyhedron were the same as the naming of the corresponding fullerene.

The results of a snub cube were summarized in Fig. 6 with polyhedron nets and planar graphs. The three patterns listed upper part (Figs. 6a, 6b, and 6c) represent the succeeded cases and the rest do the failed cases. The peelings from a square were two patterns and those from a triangle were four. Chirality affects the result in this case. Comparison of Figs. 6a and 6d shows the influence in the cases starting from

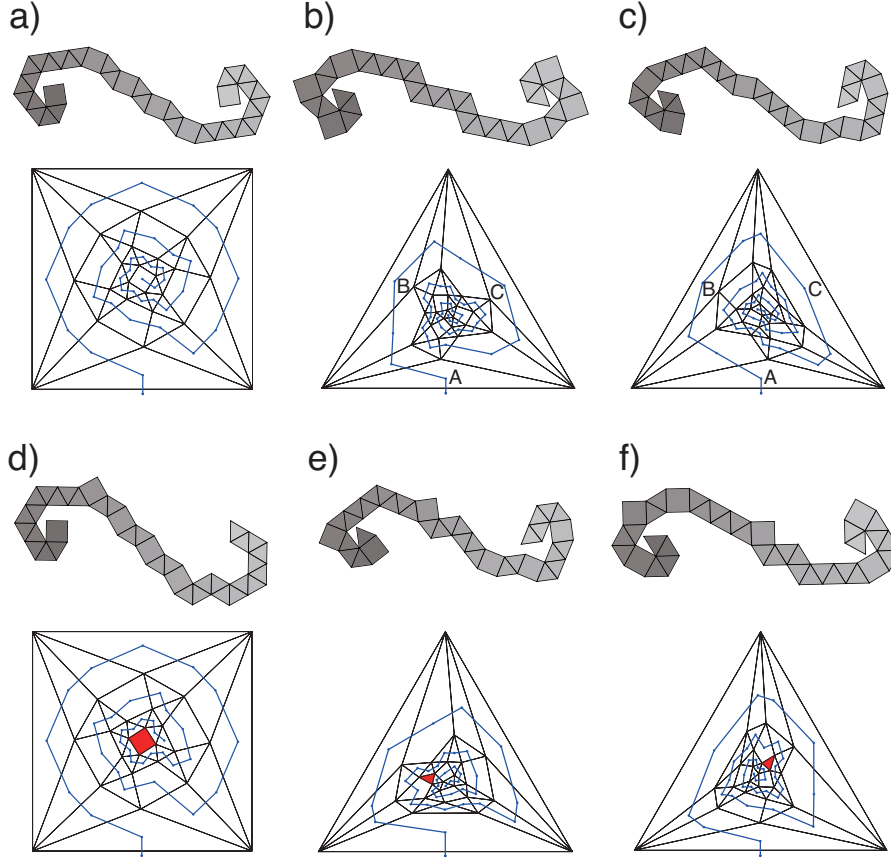


Figure 6. Comparisons between a polyhedron net and a planar graph of a snub cube $\{3, 3, 3, 3, 4\}$.

a square. In the cases of left-handed person, unfolding was succeeded in Fig. 6d and failed in Figs. 6a. There are thirteen triangles between first and second squares for the peelable cases. On the other hand, the effect appeared different way in the cases starting from a regular triangle. The results were classified into 4 types as illustrated in Figs. 6b, 6c, 6e, and 6f.

The planar graph of Fig. 6b has three-fold rotational symmetry. Therefore, the any choice among A, B, or C in Fig. 6b as F_2 results in the same. Furthermore, the it is peelable for left-handed person. On the other hand, the choice among A, B, or C in Fig. 6c reaches different result. Selecting B and C as F_2 in Fig. 6c correspond to Fig. 6e and Fig. 6f, respectively.

Although we obtained seven patterns for a truncated dodecahedrons $\{3, 10, 10\}$, as shown in Fig. 7, the basic ones were only the first three (Figs. 7a, b, and c). This result is the example of impossible polyhedrons.

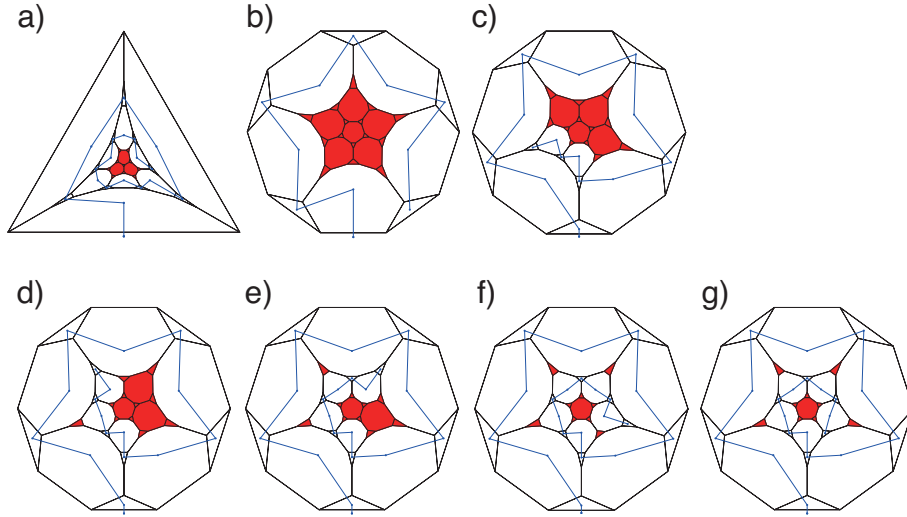


Figure 7. Summary of the peelings of truncated dodecahedrons $\{3, 10, 10\}$.

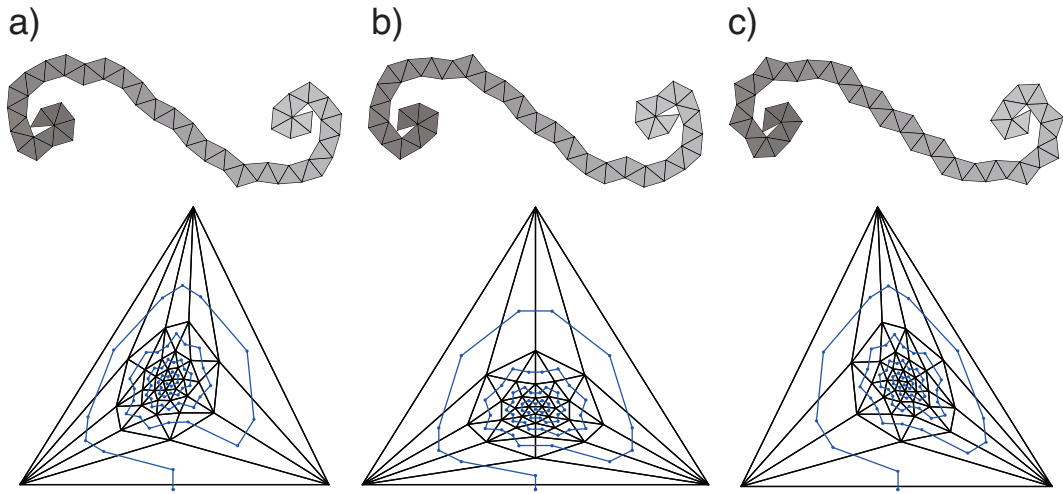


Figure 8. Pentakis Dodecahedron $[5, 6, 6]$

Figure 7 shows. Truncated Dodecahedrons $\{3, 10, 10\}$,

3.4. Catalan Solids

- Nine polyhedrons succeeded in the apple peel unfolding.

Pentakis Dodecahedron $[5, 6, 6]$, perfect Fig. 8

Tetrakis Hexahedron $[4, 6, 6]$ possible Fig. 9

Great Rhombicuboctahedron $[4, 6, 8]$, impossible Fig. 10

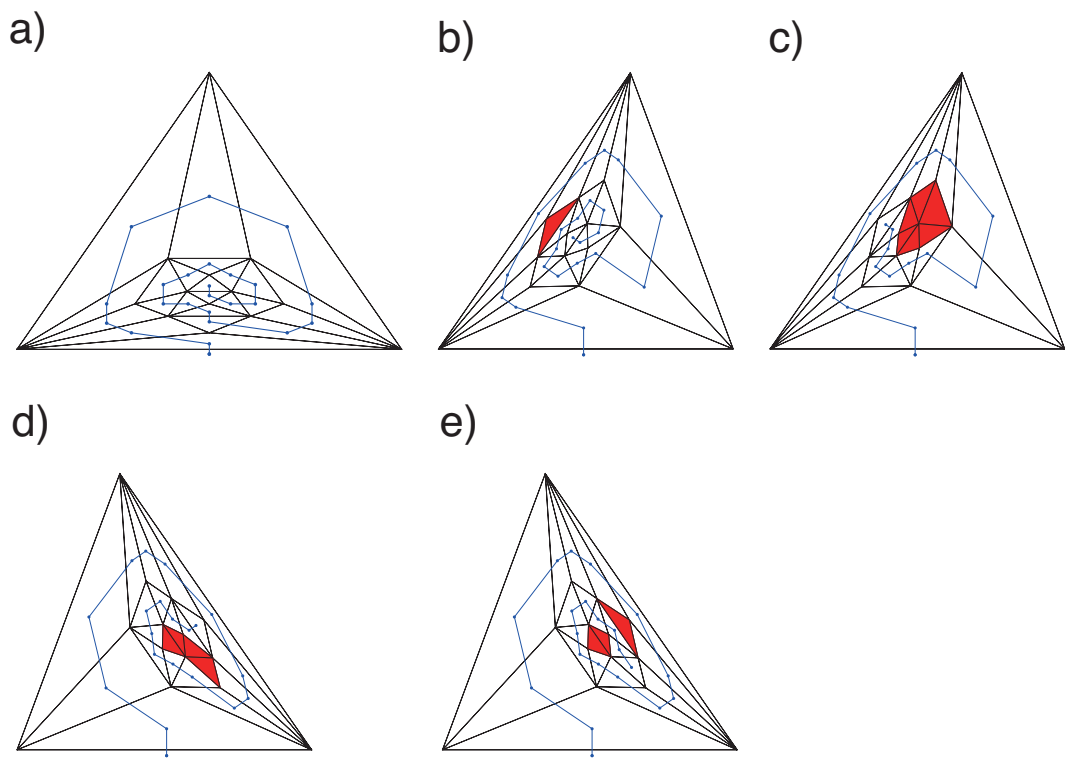


Figure 9. Tetrakis Hexahedron [4, 6, 6]

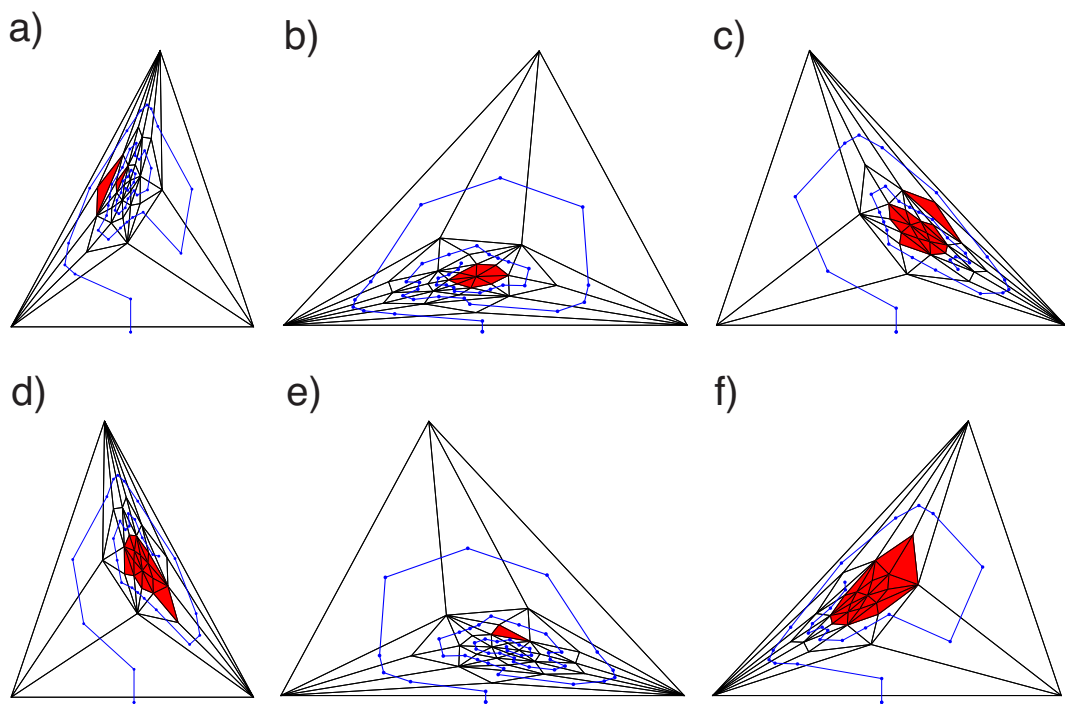


Figure 10. Great Rhombicuboctahedron [4,6,8].

4. Discussion

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An unnumbered section, e.g. `\section*{Acknowledgements}`, may be used for thanks, etc. if required and included *in the non-anonymous version* before any Notes or References.

Disclosure statement

The authors declare that they have no conflicts of interest concerning this article.

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